AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I061*

Team Name: *Gryffindor*

Domain: *Energy, Mobility and Smart Infrastructure*

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Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Innovation & Uniqueness
6. Cost for Product
7. Use Cases
8. Impacts and Benefits
9. Sustainable Development Goals
10. Individual Member Contribution

Problem Statement

To Create Efficient Utilization and Accessibility of EV Charging Infrastructure in India

Problem Understanding

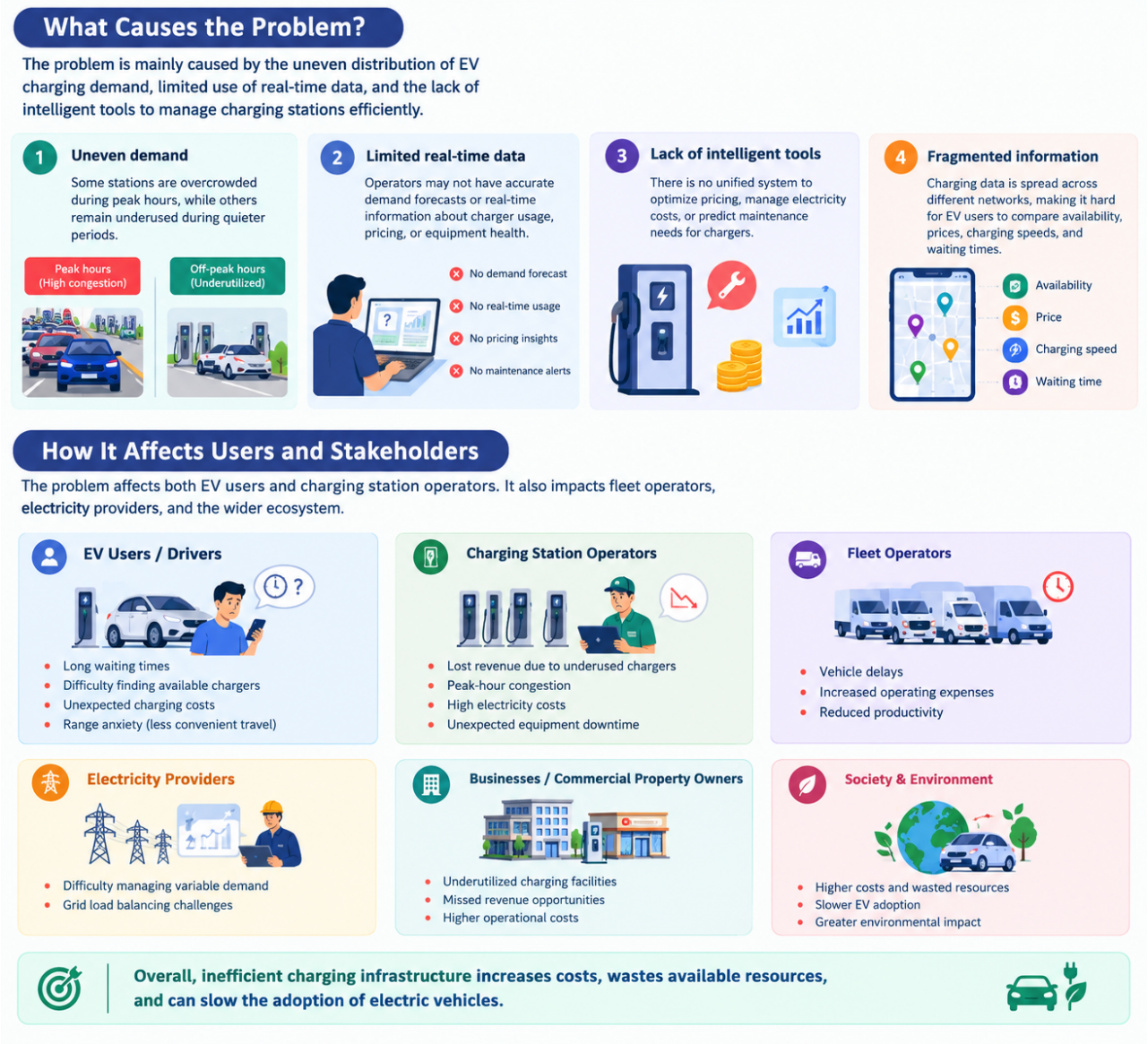


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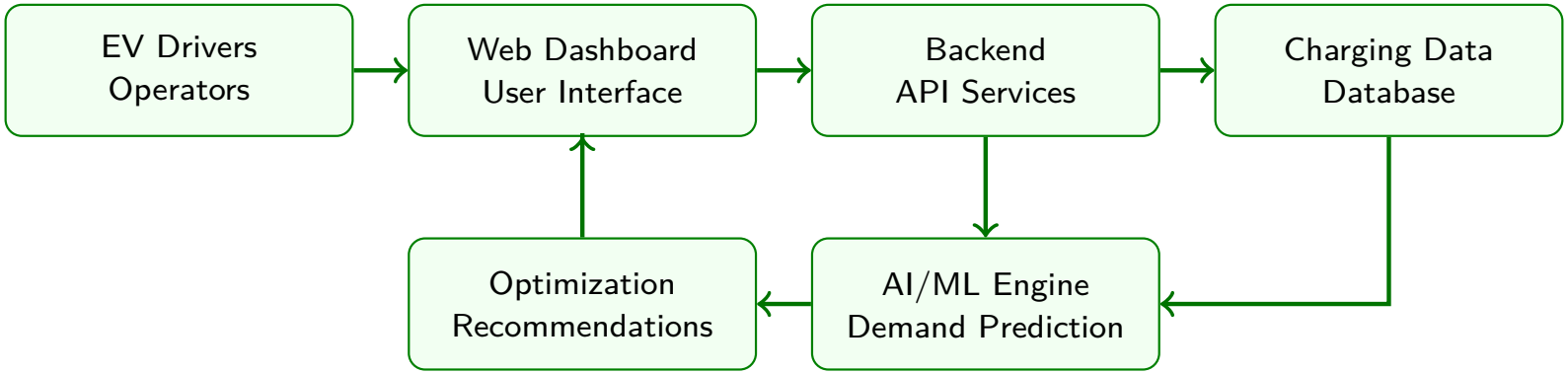
Existing Gaps & Challenges

- **Limited Demand Prediction and Charger Utilization:**
  Existing charging stations may experience peak-hour congestion and low utilization during off-peak periods. Limited demand forecasting makes it difficult for operators to balance usage and plan capacity effectively.
- **Fragmented Information and Lack of Smart Recommendations:**
  Information about charger availability, pricing, charging speed, and waiting time may be spread across different networks. Users can find it difficult to compare stations, while operators may lack integrated tools for pricing optimization, performance monitoring, and maintenance planning.
- **Main Unsolved Challenge:**
  There is a need for a unified, data-driven platform that predicts charging demand, improves station utilization, and recommends suitable chargers to EV users while balancing convenience, operating costs, and operator profitability.

Our Proposed Solution

- **Core Idea: AI-Powered EV Charging Optimization Platform**
  A smart platform that connects EV drivers with charging station operators to improve charger utilization, reduce waiting time, and increase operational profitability.
- **How the Solution Works**
  The platform analyzes charger availability, historical usage patterns, demand trends, charging prices, and waiting times. AI-based demand prediction identifies peak and off-peak periods, while the recommendation system helps drivers choose suitable stations based on distance, availability, cost, and charging speed. Operators receive insights to improve pricing, capacity planning, and maintenance decisions.
- **Main Value Delivered**
  EV users benefit from faster, more convenient, and cost-effective charging decisions. Charging operators can reduce underutilization, manage peak-hour congestion, and make better business decisions. The platform aims to improve infrastructure efficiency, reduce operational waste, and support wider EV adoption.

Technical Approach



System Workflow: The platform collects charging station data, including availability, pricing, usage, and waiting times. The backend stores and processes this information, while the AI/ML engine predicts demand and identifies utilization patterns. The optimization module recommends suitable charging stations to EV users and provides operators with insights for pricing, capacity planning, and maintenance.

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Tech Stack:

Frontend: HTML, CSS, JavaScript
Backend: Python, Flask / FastAPI
Database: SQLite / MySQL
AI/ML, APIs, or Hardware: Scikit-learn, Pandas, NumPy; simulated or public charging data

Innovation & Uniqueness

- **What Is New or Different?**
  Combines AI-based charging demand prediction, station utilization analysis, and driver recommendations in a single platform. It helps operators identify peak and off-peak periods while guiding users toward suitable chargers based on distance, price, availability, charging speed, and estimated waiting time.
- **Why Is This Approach Suited to the Problem?**
  EV charging demand changes by location and time. By analyzing historical usage and available station data, the platform can forecast demand, highlight underutilized chargers, and help operators make better pricing, capacity, and maintenance decisions. Recommendations can also help drivers avoid congestion and unnecessary detours.
- **Key Differentiator**
  A dual-benefit approach that optimizes both the operator's business performance and the driver's charging experience. Instead of only locating charging stations, it aims to make the charging network more efficient through predictive insights and data-driven recommendations.

Feasibility & Scalability

- **Feasibility:**
  A prototype can be developed using Python, machine learning, a database, and a web-based dashboard. Simulated or publicly available charging data can be used to predict demand, estimate charger utilization, and recommend suitable stations based on availability, distance, price, and waiting time. Real-time station integration can be added in later stages.

- **Scalability:**
  The platform can initially be tested with a small number of charging stations in one city and later expanded to multiple cities and charging networks. A modular architecture, cloud hosting, and API integration can support more users, stations, and data sources.

- **Risks and Mitigation:**
  Limited data may affect prediction quality; this can be approached through data validation, simulated datasets, and continuous model evaluation. Differences between charging networks may complicate integration, so standard APIs and phased partnerships can be used. Privacy and cybersecurity risks can be reduced through data minimization, access controls, and secure communication.

Cost for Product (Optional)

- **Development or Hardware Cost:**
  Rs 5,000–Rs 15,000 for prototype development, testing, and any necessary development resources. No dedicated charging hardware is required.
- **Hosting or Maintenance Cost:**
  Rs 0–Rs 2,000 per month using free-tier cloud services or a low-cost hosting plan, depending on usage and storage requirements.
- **Estimated Total:**
  Rs 5,000–Rs 17,000 for initial development and the first month of hosting, assuming the team uses existing laptops and open-source software.

Use Cases

- **Use Case 1: EV Drivers – Smart Charger Recommendation**
  An EV driver needs to charge their vehicle but wants to avoid long queues and high costs. The platform recommends suitable charging stations based on distance, availability, charging price, charging speed, and estimated waiting time, helping the driver make an informed decision.

- **Use Case 2: Charging Station Operators – Demand and Profit Optimization**
  A charging station operator experiences low usage during off-peak hours and congestion during peak periods. The platform predicts demand using historical and available real-time data, identifies underutilized chargers, and provides insights for pricing, capacity planning, and maintenance decisions.

- **Use Case 3: Fleet Operators – Cost and Downtime Reduction**
  A delivery or taxi fleet needs to charge multiple EVs while maintaining its schedule. The platform helps identify suitable charging stations and plan charging based on availability, cost, and expected waiting time, reducing avoidable delays and improving fleet utilization.

Impacts and Benefits

- **Benefits to Target Users:**
  EV drivers can identify suitable charging stations based on distance, price, availability, charging speed, and estimated waiting time. Charging station operators gain demand insights to improve charger utilization, pricing decisions, and maintenance planning. Fleet operators can plan charging more efficiently to reduce avoidable delays and operating costs.
- **Social, Economic, and Environmental Impact:**
  The platform can improve access to charging information, reduce wasted charging capacity, and support more efficient use of electricity and infrastructure. Better charging experiences may encourage EV adoption, while improved operational efficiency can benefit businesses and contribute to more sustainable urban mobility.
- **Expected Measurable Outcomes:**
  Evaluate the prototype using average predicted waiting time, charger utilization rate, recommendation accuracy, estimated charging cost, and demand prediction error. Compare results against a baseline without optimization to measure potential improvements in user convenience and station efficiency.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goals	Connection
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Uses AI and data analytics to improve charging infrastructure
SDG 11	Target 11.2	Sustainable Cities and Communities	Helps make urban transportation more accessible and efficient

Individual Member Contribution

Member	Contribution or responsibility
Sarvesh S	Worked on presentation preparation, feasibility analysis, and identifying the project's impacts and benefits.
Dilipkumar A	Led problem identification, project planning, and defining the proposed solution.
Abhijith S	Researched existing EV charging systems and analyzed market needs and competitor solutions.
Muhammad Rayyan K A	Contributed across research, solution design, technical planning, feasibility analysis, and presentation development.

Yukti ID

IR2026-1051014